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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S3 (R,S) / S3 (WP) (R,S) / S1 (PT) (S,FE) Examination November 2024 (2019 Scheme)

Course Code: EET205

Course Name: ANALOG ELECTRONICS

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions. Each question carries 3 marks

Marks

- 1 What is meant by thermal runaway? What all parameters affect the stability of operating point of BJT? (3)
- 2 How bias compensation is done in BJT using thermistors? (3)
- 3 The data sheet of an N- channel JFET gives the following details. $I_{DSS} = 9\text{mA}$ and pinch off voltage, $V_P = -4.5\text{V}$. i) At what value of gate to source voltage V_{GS} , drain current I_D will be equal to 3mA ? (3)
- 4 Compare the features of FET and BJT. (3)
- 5 Differentiate between positive and negative feedback circuits. Why negative feedback is utilized in amplifiers? (3)
- 6 State and explain the Barkhausen criterion for sustained oscillation? (3)
- 7 What are the ideal characteristics of operational Amplifier? (3)
- 8 What is a Voltage follower circuit using OP - Amp? (3)
- 9 Draw a practical integrator and explain the purpose of feedback resistance in practical integrator. (3)
- 10 Design a comparator using OP-Amp that compares a sinusoidal signal of 5V peak with a fixed dc voltage of 2.5V . Draw corresponding waveforms. (3)

PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1

- 11 (a) Derive the stability factor of a voltage divider biasing circuit of a BJT amplifier. (6)
- (b) Obtain the operating point set by the voltage divider bias circuit for an NPN CE transistor with $\beta = 80$ and $V_{BE} = 0.7\text{V}$. Given $V_{CC} = 16\text{V}$, $R_1 = 62\text{k}\Omega$, $R_2 = 9.1\text{k}\Omega$, $R_C = 3.9\text{k}\Omega$ and $R_E = 680\Omega$. Also calculate stability factor, S. (8)

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- 12 (a) Derive the expressions for voltage gain, current gain and input impedance for BJT in common emitter configuration using h-parameter model. (6)
- (b) A transistor in Common Emitter configuration has the following set of h parameters. $h_{ie}=1700\Omega$, $h_{re}=1.3 \times 10^{-4}$, $h_{fe} = 50$, $h_{oe}= 6 \times 10^{-6} \Omega$. If the ac load resistance r_L seen by the transistor is $2k\Omega$, calculate (i) input impedance (ii) current gain and (iii) voltage gain. (8)

Module 2

- 13 (a) Draw and explain the drain and transfer characteristics of JFET. What is meant by I_{DSS} and Pinch-off voltage, V_P ? (8)
- (b) Explain the construction and working principle of E-MOSFET (6)
- 14 (a) Draw and explain the voltage divider biasing method of JFET. (6)
- (b) In an n- channel JFET, biased by potential divider method, it is desired to fix the operating point at $I_D = 3mA$ and $V_{DS} = 8V$. If $V_{DD} = 20V$, $R_1 = 1M\Omega$, $R_2 = 500k\Omega$, find the value of R_S . The parameters of JFET are $I_{DSS} = 9mA$ and $V_P = -5V$. (8)

Module 3

- 15 (a) What is the basic principle of R-C phase shift Oscillators? Design an R-C phase shift oscillator to oscillate at 500 Hz. (8)
- (b) Draw and explain Colpitts oscillator and derive its frequency of oscillation. (6)
- 16 (a) With necessary diagrams explain the working of class A transformer coupled amplifier and obtain its maximum conversion efficiency. (8)
- (b) Explain different schemes of coupling in multistage amplifiers. Compare their merits and demerits. (6)

Module 4

- 17 (a) Define the following terms of OP-Amp (8)
- (i) CMRR (ii) Slew rate (iii) Input bias current (iv) Output offset voltage
- (b) Determine the gain and maximum operating frequency of an inverting amplifier with feedback resistance $R_f = 100k\Omega$ and input resistance $R_i = 10k\Omega$ for a sinusoidal input signal of 1 volt peak. (take slew rate as $0.5V/\mu s$). (6)
- 18 (a) Design an OP-Amp circuit to get the output according to the given expression $V_o = -[0.2V_1 + 2V_2 + V_3]$ where V_1 , V_2 and V_3 are the inputs to OP-Amp. (6)
- (b) What is an instrumentation amplifier? Derive the expression for output voltage of an instrumentation amplifier. (8)

Module 5

- 19 (a) Draw and explain the operation of a square wave generator using OP-Amp (8)
Derive the equation for its output frequency.
- (b) In an inverting Schmitt trigger circuit, $R_1 = 40 \text{ k}\Omega$, $R_2 = 10 \text{ k}\Omega$ (feedback factor $\beta = R_2/(R_1 + R_2)$), reference voltage, $V_{\text{ref}} = 0 \text{ V}$, input voltage, $V_i = 12 \text{ V}_{\text{pp}}$ (peak-to-peak) sine wave and saturation voltage are $\pm 15 \text{ V}$. Determine threshold voltages V_{UT} and V_{LT} and also draw input and output waveforms. (6)
- 20 (a) With the help of internal function diagram of IC555, explain its operation as astable multivibrator. Derive the expression for frequency of oscillation. (8)
- (b) Design a monostable multivibrator using IC555 to produce a time delay of 10 ms. (6)
